AI ENGINEERING / OPERATIONS / FINANCE / CASE STUDY

CASE STUDY

09

DELIVERY MATCHED INVOICING

A synthetic food distributor tests whether invoices can follow what actually passed goods in.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A synthetic distributor receives chilled and ambient deliveries at several depots every day. Suppliers invoice in full for the order even when pallets are rejected on temperature or quality at the door. Credits depend on someone noticing and chasing weeks later.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Goods in supervisor**	Wants rejection records to be used rather than lost in a handheld device or a paper note.
02	**Accounts payable lead**	Needs invoices paid for the right quantity without causing supplier disputes.
03	**Buyer**	Owns the supplier relationship and decides how firmly each credit is pursued.

SCOPE AND HONEST LIMITS

- Suppliers and deliveries are synthetic so real handheld and paper quirks are not captured.
- No recovered value is claimed; the plan is to measure matching on labelled data.
- Supplier credit terms and dispute behaviour are not modelled.
- Photo evidence and verbal agreements at the dock are out of scope.

CURRENT WORKFLOW AND FAILURE POINTS

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PROPOSED RESEARCH / NOT YET VERIFIED

HOW THE WORK RUNS TODAY

1 TODAY

Delivery arrives

Goods in checks temperature and condition and records accepted and rejected quantities.

2 TODAY

Rejection noted

The reason is written on the delivery note or typed into a handheld with free text.

3 TODAY

Invoice arrives

The supplier sends an invoice for the full ordered quantity by email or portal.

4 TODAY

Two way check

Accounts payable compares the invoice to the purchase order and not to what was received.

5 TODAY

Payment approved

The invoice is approved on price and quantity ordered and enters the payment run.

6 TODAY

Credit chased later

A buyer may notice the rejection weeks on and email the supplier for a credit note.

WHERE IT BREAKS

Overpayment	Rejected goods are paid for when nobody links the receipt to the invoice.
Lost evidence	Rejection reasons and temperature readings are scattered across notes and devices.
Late credits	Chasing weeks after the delivery weakens the claim and strains the supplier relationship.
Manual effort	Matching partial deliveries by hand takes time that is rarely available.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

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PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW

1 SYSTEM
Capture receipt
Goods received notes and rejection records are stored as structured lines with reasons.

→

2 SYSTEM
Ingest invoice
The invoice is parsed and each line is linked to the order and its delivery.

→

3 SYSTEM
Match to received
Billed quantities are compared with accepted quantities and substitutions are flagged.

↓

4 SYSTEM
Draft credit request
For each rejected line a factual request is drafted citing the record and reading.

→

5 PERSON
Buyer approval
A person edits and approves or declines every draft before it goes to a supplier.

→

6 PERSON
Pay and reconcile
Accounts payable pays the agreed amount and tracks the credit note until it arrives.

SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Whether to send a credit request	BUYER	It affects a supplier relationship that only the buyer can judge.
Whether to hold or part pay an invoice	ACCOUNTS PAYABLE LEAD	Payment terms and cash timing are a finance decision.
Whether a rejection was valid	GOODS IN SUPERVISOR	Only the person at the dock knows the condition of the goods.

ARCHITECTURE, DATA AND CONTROLS

09

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM ARCHITECTURE

1	**Receipt capture**	Structured goods received notes with temperature readings and rejection reasons by line.
2	**Invoice ingestion**	Parses supplier invoices into lines and ties them to purchase orders and deliveries.
3	**Matching engine**	Reconciles ordered and received and billed quantities with units and substitutions handled.
4	**Credit drafting**	Produces line level credit request drafts with the evidence that supports each one.
5	**Approval and tracking**	Queue for buyer decisions that logs approvals and follows credit notes to completion.

CORE DATA ENTITIES

- **Supplier**
- **Purchase order**
- **Goods received note**
- **Rejection record**
- **Invoice line**
- **Credit request**

RISKS AND THE CONTROL FOR EACH

Credit requested for accepted goods →	Every draft must cite a rejection record and a buyer approves it before sending.
Missing receipt treated as rejection →	Lines without a received record go to a person and never default to a credit.
Unit or pack size mismatch →	Unit conversions are explicit and failed conversions route the line to review.
Automatic supplier contact →	The system has no ability to send a request or hold payment without a person.

EVALUATION AND DELIVERY PLAN

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PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Credit recall	Share of injected synthetic rejections that appear in a credit draft.
Credit precision	Share of drafts that a reviewer would send without removing the line.
Match rate	Share of invoice lines matched to a delivery line without human help.
Approver edit rate	Share of drafts a buyer changes before approval.
Time to draft	Elapsed time from invoice receipt to a draft ready for review.

TEST PLAN

1. **Reject half a pallet in the synthetic data and confirm the draft credits exactly that quantity.**
2. **Deliver in two parts and confirm the invoice matches across both receipts.**
3. **Substitute a product and confirm the line is flagged and not credited.**
4. **Remove the rejection note and confirm the line goes to a person with no credit drafted.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree which rejection reasons and suppliers are in scope.	Walk the dock process and map how rejections are recorded.	Design the match rules and the credit request wording.	Build the matcher and approval queue on synthetic deliveries.	Run beside the current process with buyers approving each draft.
DECISION Which rejection types count as creditable	**DECISION** Where the source of truth for receipts lives	**DECISION** How tone and detail are set for suppliers	**DECISION** Whether precision is high enough for review	**DECISION** Whether to widen to more depots

DELIVERABLES

- **Synthetic orders receipts and invoices dataset**
- **Matching rule catalogue**
- **Credit request draft templates**
- **Buyer approval workflow design**
- **Evaluation report with false credit cases**